

MELEUZ, Russian Federation

WMO: 289250

Lat: 52.9302N

Lon: 55.9174E

Elev: 180

StdP: 99.18

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-28.8	-34.4	0.1	-31.7	-31.4	0.2	-28.7	11.8	-6.2	9.8	-5.6	0.5	200	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.9	20.5	29.9	19.7	28.2	19.0	22.5	29.2	21.2	27.5	19.9	26.3	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.1	26.6	18.9	14.0	25.2	17.6	12.9	23.3	67.2	29.3	62.2	27.7	57.8	26.2	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.9	7.6	6.7		DB	-37.6	35.0	4.1	2.2	-40.5	36.5	-42.9	37.8	-45.2	39.0	-48.2	40.6
				WB	-36.7	23.7	4.8	2.1	-40.2	25.2	-43.0	26.4	-45.7	27.6	-49.3	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-12.2	-11.8	-6.0	5.2	13.3	18.4	20.4	17.9	12.4	5.3	-4.9	-11.4
	DBStd	13.47	7.54	7.88	5.90	6.44	4.76	4.12	3.32	3.93	4.72	4.92	6.75	8.34
	HDD10.0	3291	689	611	497	169	24	1	0	0	27	162	448	662
	HDD18.3	5435	947	844	756	393	164	51	18	54	184	405	698	921
	CDD10.0	1089	0	0	0	27	128	252	321	246	99	16	0	0
	CDD18.3	191	0	0	0	1	9	52	81	42	7	0	0	0
	CDH23.3	2159	0	0	0	14	156	589	822	474	104	0	0	0
	CDH26.7	730	0	0	0	1	28	212	281	176	33	0	0	0
Wind	WSAvg	2.4	2.5	2.8	2.4	2.3	2.7	2.3	2.1	2.0	2.1	2.6	2.5	2.6
Precipitation	PrecAvg	455	37	28	35	26	32	55	42	41	30	48	37	44
	PrecMax	587	60	58	86	61	98	102	80	78	47	98	80	106
	PrecMin	295	9	9	4	1	2	25	2	6	9	8	4	13
	PrecStd	82	17	14	25	19	24	25	24	19	14	26	23	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.6	7.1	25.2	29.6	35.0	33.8	34.1	31.0	21.5	8.9	1.7
		MCWB	0.3	0.7	3.8	15.4	16.3	20.7	21.5	21.9	17.4	13.9	7.8	0.7
	2%	DB	0.1	0.8	3.9	22.1	27.6	31.2	31.5	31.1	25.8	18.3	5.8	0.8
		MCWB	-0.6	-0.3	1.6	12.9	16.1	19.9	20.9	20.5	15.1	10.9	4.8	0.1
	5%	DB	-1.0	0.0	2.7	19.2	25.3	29.0	29.8	28.3	23.3	15.5	4.2	0.2
		MCWB	-1.6	-0.7	1.2	11.4	15.2	19.4	20.4	19.2	14.8	9.8	3.1	-0.4
	10%	DB	-2.3	-1.2	1.7	15.7	22.6	26.9	28.1	25.7	20.8	13.0	2.8	-0.7
		MCWB	-3.0	-2.0	0.5	9.7	14.2	18.8	19.9	18.1	13.6	9.1	1.6	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.9	4.2	16.3	19.5	23.2	24.3	23.5	18.4	15.3	7.7	1.1
		MCDB	1.0	1.9	6.2	24.6	25.9	30.3	30.5	31.6	27.9	19.9	8.7	1.6
	2%	WB	-0.5	0.0	2.3	13.5	17.0	21.9	22.8	21.7	16.5	12.5	4.8	0.4
		MCDB	0.0	0.6	3.5	20.6	25.1	28.2	29.2	28.3	24.1	16.3	5.4	0.8
	5%	WB	-1.5	-0.7	1.4	11.6	16.0	20.6	21.4	19.9	15.2	10.7	3.4	-0.3
		MCDB	-1.0	0.0	2.5	18.1	23.7	27.0	27.7	25.9	21.6	14.2	4.1	0.2
	10%	WB	-3.0	-2.0	0.5	9.9	14.7	19.2	20.2	18.7	14.1	9.1	1.8	-1.3
		MCDB	-2.4	-1.3	1.7	15.2	21.4	25.3	26.6	24.8	19.9	12.8	2.5	-0.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.6	8.7	10.3	11.6	11.6	11.0	11.5	11.0	8.1	6.0	7.5
		MCDBR	5.1	6.0	6.8	15.5	16.1	15.0	14.1	15.6	15.7	13.6	5.9	5.9
		MCWBR	4.9	5.5	5.5	7.9	7.2	6.8	6.0	6.6	7.4	7.9	5.1	5.8
	5% WB	MCDBR	5.2	5.6	5.8	14.6	14.3	13.4	13.0	13.8	14.3	12.2	4.5	5.5
		MCWBR	5.1	5.4	4.9	8.0	7.4	7.3	6.7	7.0	7.6	8.0	4.4	5.6
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.379	0.427	0.407	0.399	0.411	0.432	0.381	0.353	0.326	0.298	
	taud	2.050	2.009	1.931	2.045	2.245	2.327	2.294	2.215	2.324	2.303	2.173	2.033	
	Ebn at Noon	638	745	779	798	842	853	833	788	788	725	610	553	
	Edh at Noon	76	109	146	149	129	120	122	125	99	82	68	65	
All-Sky Solar Radiation	RadAvg	1.04	2.13	3.54	4.53	5.54	6.11	5.94	4.83	3.26	1.61	0.74	0.67	
	RadStd	0.12	0.18	0.36	0.45	0.48	0.66	0.50	0.44	0.40	0.32	0.16	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page